

FD-RLPO: Feature Domain-based Reinforcement Learning Framework for Portfolio Optimization

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Abstract—Portfolio optimization is a critical issue in finance. In the past decade, reinforcement learning has advanced considerably in this area owing to its outstanding ability to solve complex sequential decision-making problems. Nevertheless, existing methods still have great room for improvement in fully considering the dynamic fluctuations of intra-domain and inter-domain features in financial markets. To address this issue, we propose FD-RLPO, a Feature Domain-based Reinforcement Learning Framework for Portfolio Optimization, which integrates Prediction and Relation Modules while leveraging reinforcement learning agents for adaptive strategy adjustment. In particular, the Prediction Module utilizes an inversion mechanism within a Transformer architecture to effectively extract sequential fluctuations in intra-domain features. Meanwhile, the Relation Module employs a self-supervised approach to capture inter-domain relational distribution patterns, mitigating the impact of data noise on reconstructed asset relationships. Ultimately, the Decision Module constructs a comprehensive representation of the market feature domain by synthesizing the outputs of the preceding modules and leveraging the Actor-Critic Network to achieve portfolio optimization. The simultaneous perception of both intra-domain and inter-domain features enables FD-RLPO to dynamically adjust to the volatility of financial markets and increase returns. Experiments on the CSI-300 and NASDAQ-100 datasets demonstrate that FD-RLPO outperforms previous state-of-the-art methods in key metrics, including the Annualized Rate of Return (ARR) and Annualized Sharpe Ratio (ASR). Ablation studies further validate the effectiveness of each component.

Index Terms—portfolio optimization, reinforcement learning, feature domain, transformer, self-supervised, inversion mechanism

I. INTRODUCTION

Portfolio optimization is defined as reallocating diversified financial assets to maximize returns. To achieve this, various strategies have been proposed to balance risks and returns [1]. Reinforcement learning (RL) has emerged as a key approach due to its flexibility in incorporating constraints such as transaction costs while optimizing returns. However, in recent years, as market complexity has increased, the volatility of financial market properties—such as time-series volatility and inter-asset relationships—has risen, making decision-making even more challenging. As a core source of market dynamics, the behavior of feature domains influences market volatility, which in turn affects risk and return.

Stock feature domains—such as price, volume, technical indicators and correlations across different stocks—change

dynamically over time, forming distinct feature domains at each timestamp. Comprehensive modeling requires considering two aspects: intra-domain features, which capture intrinsic patterns within individual series (e.g., opening prices), and inter-domain features, which reflect correlations among stocks or industries. First, previous work [2] utilized a fractal feature selection method in conjunction with support vector machines to capture intra-domain relationships in stocks. However, the static nature of fractal feature selection limited the model's performance. With the emergence of the Transformer architecture, static feature selection can be avoided. [3] utilizes the self-attention mechanism, enabling the model to weigh the importance of each input feature relative to others. Additionally, leveraging self-attention distillation improves inference speed. Even so, Transformer-based methods typically focus on feature mining along the temporal dimension to create time tokens. These approaches neglect the consistency of feature sequences, leading to inadequate handling of the inherent volatility of stock intra-domain features.

Secondly, financial assets in portfolios exhibit complex inter-domain relationships that may vary rapidly over time. Previous research [4] [5] has made great contributions in this field. For instance, [5] employs Dynamic Time Warping (DTW) to construct lead-lag graphs quantifying inter-stock relationships, but its reliance on a predefined graph structure incurs high computational costs and noise sensitivity. Similarly, [6] employs self-attention to model nonlinear asset correlations and uses deterministic policy gradient-based recurrent reinforcement learning for decision-making, relying heavily on high-quality data. Therefore, it is crucial to effectively capture the inter-domain correlations by reducing interference from data noise for portfolio optimization.

In this work, we propose a feature domain-based architecture with three modules designed to simultaneously capture both intra- and inter-domain features to diminish the impact of market volatility and enhance returns. We innovatively design a Prediction Module based on a Transformer architecture with an inversion mechanism, constructing tokens centered on variables rather than timestamps. Additionally, in the Relation Module, we introduce a self-supervised approach to reconstruct data, supported by subsequent loss backpropagation, which alleviates the impact of data noise and reveals better

inter-domain relationships. A multi-head attention mechanism then integrates these features into a rich market state space. Subsequently, in the Prediction Module, reinforcement learning is employed to train an Actor-Critic framework within the state space, facilitating the optimization of returns and thereby ensuring positive gains. The contributions of this paper are summarized as follows:

- We propose FD-RLPO, a novel framework that simultaneously captures intra- and inter-domain features to address financial data fluctuation challenges.
- We develop a Prediction Module that novelly applies a feature-centered method utilizing an inversion mechanism to achieve effective intra-domain feature extraction and mitigate intra-sequence volatility. For the Relation Module, we apply a self-supervised Transformer to facilitate inter-domain stock relational learning, which reduces the impact of noise and enhances feature consistency. Furthermore, we employ an SAC network as the backbone to attain a strengthened balance of risk and return.
- Our proposed model achieves significant improvements over previous methods. On the NASDAQ-100 dataset, it achieves an average improvement of 9.7% in ARR, 0.89 in ASR, and a substantial 4.69% reduction in Maximum Drawdown (MDD) compared to the baselines. Ablation studies further confirm the efficacy of the critical components of FD-RLPO.

II. RELATED WORKS

A. Portfolio Management Methods

Recently, reinforcement learning (RL) models have been employed for portfolio optimization by training agents to optimize reward functions and enhance trading behavior using an end-to-end approach that directly maps market data to trading actions [7]. For example, Proximal Policy Optimization (PPO) [8] incorporates trust region constraints and a clipped objective function to address issues of slow convergence and instability in policy gradient methods. However, it lacks the ability to effectively manage noise in continuous, high-dimensional state spaces. Deep Q-Networks, such as NCADRL4TEA [9], optimize decision-making in multi-agent systems by leveraging a leaving policy for global exploration. Furthermore, Deep Deterministic Policy Gradient (DDPG) [10] employs deterministic policy gradients and Q-function estimation to address challenges in learning policies for continuous action spaces. However, DDPG exhibits suboptimal training stability, particularly under noisy conditions, resulting in inconsistent performance.

In this paper, we implement the Soft Actor-Critic (SAC) [11] as the Decision Module. SAC optimizes a weighted objective that balances reward and policy entropy for enhanced robustness. Its off-policy update efficiently utilizes historical data to mitigate real-time noise. Additionally, the adaptive entropy coefficient dynamically balances exploration and exploitation, enabling SAC to flexibly adapt to high-noise and non-stationary environments.

B. Feature Domain Capture Techniques

Deep learning models have been widely used in recent years to extract intra-domain properties from financial data. For instance, [12] employs CNNs for feature extraction and BiLSTM networks to capture temporal dependencies, effectively handling sequence nonlinearity. However, this approach responds slowly to short-term abrupt changes. Then, Transformer-based architectures, such as FactorVAE [13] and Informer [3], emerged as another category, utilizing attention mechanisms that assign equal importance to all time steps. This enables better modeling of short-term dynamics during abrupt changes. Nevertheless, most Transformer-based methods focus on intra-domain features along the time axis, disrupting sequence consistency and underutilizing feature-centered information. To address this, we propose an inversion mechanism within a Transformer-based Prediction Module, constructing tokens along the feature dimension to enhance intra-domain representation, ensure sequence completeness, and reduce intra-sequence volatility.

Additionally, methods for capturing inter-domain features have evolved significantly. Recent models, such as [4], [12], frequently adopt union graphs to effectively capture inter-domain relationships. For instance, [14] models stock price charts as images and integrates them into neural networks for short-term forecasting, mimicking financial experts' analysis. However, this approach is noise-sensitive, especially in volatile markets, as it may amplify short-term price fluctuations. Similarly, [15] captures multidimensional interactions among short-term stocks but relies heavily on high-quality data for constructing multi-relational graphs. Therefore, as observed in [16], self-supervised learning in multi-asset scenarios effectively captures nonlinear and diverse inter-asset relationships using methods like contrastive learning. To leverage this, we propose a jointly self-supervised Relation Module that dynamically learns inter-domain features in high-noise environments.

III. METHODOLOGY

A. Overall Architecture

Complex financial market volatility raises two key challenges: preserving intra-domain feature sequence integrity and mitigating noise interference in inter-domain relationship modeling. Inspired by [17], FD-RLPO addresses these issues through a multi-module framework for effective portfolio optimization, as illustrated in Figure 1. This framework comprises three modules—Relation, Prediction, and Decision. The Relation Module employs self-supervised random masking and critic loss to reconstruct inter-stock relationships, while the Prediction Module incorporates an inversion mechanism in the encoder to extract inter-domain representations. Outputs from these modules are integrated via multi-attention into compositional representations S_d , which the Actor network then uses to compute the trading strategy function π_θ and determine the optimal trading operation.

These aforementioned modules will be elaborated in detail in the following sections.

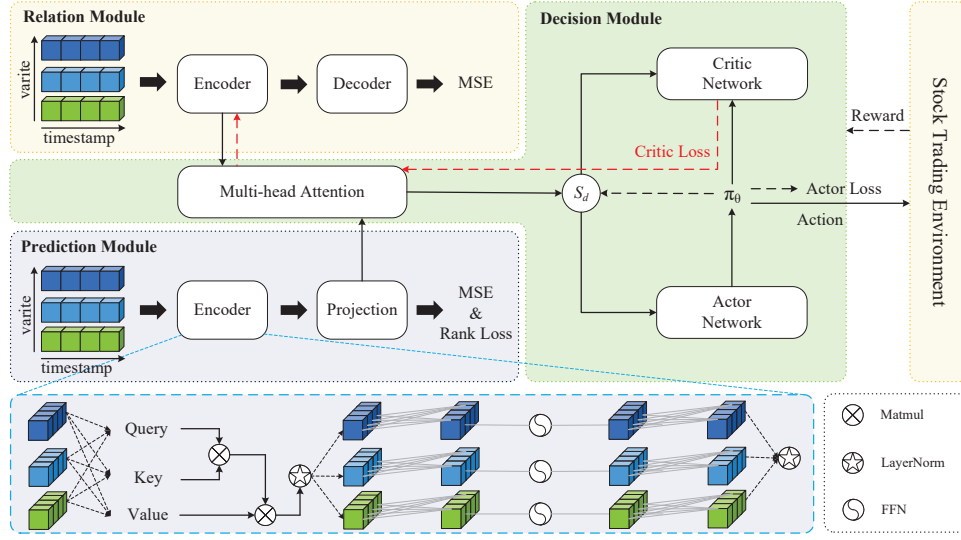


Fig. 1. Overview of FD-RLPO Architecture.

B. Relation Module

The Relation Module (a) uses self-supervised learning by randomly masking half of the data, enabling the encoder to reconstruct output representations effectively, and (b) incorporates DMH attention with multiple FFNs to enhance feature decoupling, collectively improving inter-domain feature capture and accurately representing relationships between assets.

To improve extracting valid dependency representations in inter-domain features. Our model employed Diversified Multi-head Attention (DMH-Attn), with a group of feed-forward networks (FFNs), is used instead of a single single FFN, in which each FFN individually responds to a single head in the output of the multi-head attention layer. Specifically, The stock variable series, as shown in Figure 1, enters the encoder of the Relation Module as input. The process is represented as follows.

$$X = \text{MH-Attn}(Q, K, V) + Q \quad (1)$$

$$f_i = \max(0, x_i W_{1,i} + b_{1,i}) W_{2,i} + b_{2,i} \quad (2)$$

$$\text{DMH-Attn}(Q, K, V) = \text{Concat}(f_1, \dots, f_h) + X \quad (3)$$

After passing through the standard multi-head attention (1), X will be segmented by channel dimension N as x_i , and fed into the feed-forward neural network FFN (2), merging $n f_i$ to get the final encoder output (3), providing a representation of the current relationship between stocks for subsequent Decision Modules.

Like the previously mentioned self-supervised learning [16], which treats the data as a corrupted input to the encoder and increases the accuracy of the reconstructed time series from the output of that encoder, we use self-supervised learning in our structure. At the present time step, we use the same input for the encoder and the decoder, $H_{\text{enc},t} = H_{\text{dec},t} \in \mathbb{R}^{N \times (K+N)}$. Here, N represents the number of concurrently traded stocks,

while K denotes the number of statistical measures used to capture the dynamics of the time series data. The representation $H_{\text{enc},t}$ consists of two parts: one part forms the covariance matrix, and the other part is $O_{\text{relat},t}$. For $O_{\text{relat},t}$, we randomly select half of the time series and mask their input statistics by zero during self-supervised training, but only during the training stage, the inference stage does not utilize this approach. In this process, Mean Squared Error (MSE) is used as the loss function (4), where $\hat{O}_t^{\text{relat}}$ and $\bar{O}_t^{\text{relat}}$ are the reconstructed values of the masked statistics and their ground truth.

$$L_{\text{relat}} = \frac{1}{n} \sum_n \|\hat{O}_t^{\text{relat}} - \bar{O}_t^{\text{relat}}\|^2 \quad (4)$$

It is worth noting that the Relation Module not only relies on its own loss function update but also performs parameter optimization through the subsequent gradient backpropagation of the Critic network, enabling it to better capture true inter-domain stock features.

C. Prediction Module

To address dynamic intra-domain feature fluctuations in financial data, we design an innovative inversion mechanism integrated into the Transformer architecture. Traditional Transformer models in this area treat time-series data as temporal tokens, which suffer from attention dispersion as sequence length increases, further limiting their ability to effectively capture intra-sequence information and exacerbating volatility [18]. In this setup, the inversion mechanism swaps the time and variable dimensions by applying attention to variable tokens instead of ensuring variable consistency. This way, converting the original time-attention problem into a variable-attention problem, thereby reducing the competition for attention weights across time steps. The mechanism fully exploits the parallel capabilities of multi-head attention, ensuring that feature associations across long time spans are effectively

extracted, while mitigating intra-sequence volatility. In this context, the objective of the Prediction Module is to forecast each stock's return rate over a five-day period, aligning with a typical trading week in the stock market. The return R_t is computed as (5). $P_{\text{close},t}$ is the closing price at time t , which means that the core is price prediction.

$$R_t = (P_{\text{close},t} - P_{\text{close},t-1}) / P_{\text{close},t-1} \quad (5)$$

Specifically, the time series of each stock (illustrated by distinct colors) is first tokenized to characterize the variables before being fed into the encoder. Rather than employing the attention mechanism to improve temporal dependencies as previously implemented, attention is applied to the embedded variables to mine information centered on the variables, with the entire sequence of each variable considered as an independent process. In this case, the *Query* and *Key*^T calculate the dot product of different variable sequences, and the resulting attention score reflects the strength of the interactions between variables as Z . Layer normalization (LN) is then applied to the variables rather than to the timestamps, as shown in the following (6), where k represents the number of tokens.

$$\text{LayerNorm}(Z) = \left\{ \frac{z_k - \text{Mean}(z_k)}{\sqrt{\text{Var}(z_k)}} \mid k = 1, \dots, K \right\} \quad (6)$$

We employ an encoder-only architecture, where the task of generating the predicted series is primarily handled by linear layers, a method that has been proven effective in previous studies. In this part, we utilize MSE and Rank loss as loss functions for joint optimization, as detailed below (7) (8):

$$\mathcal{L}_{\text{rank}} = \sum_{i=1}^N \sum_{j=1}^N \max(0, -(r_i - r_j)(\hat{r}_i - \hat{r}_j)) \quad (7)$$

$$\mathcal{L}_{\text{Pred}} = \frac{1}{N} \|r_i - \hat{r}_i\|^2 + \alpha \mathcal{L}_{\text{rank}} \quad (8)$$

r_i, r_j denote the true returns of different stocks, while $\hat{r}_i$ and $\hat{r}_j$ represent the predicted returns. The purpose of $\mathcal{L}_{\text{rank}}$ is to ensure that the predicted ranking closely aligns with the original ranking. Together with the mean squared error (MSE), these form the joint composition of $\mathcal{L}_{\text{Pred}}$. This module utilizes the output features of the decoder as one of the inputs to the Decision Module.

D. Decision Module

Previously, intra- and inter-domain representations of features from two modules were integrated via a multi-attention mechanism resulted in a comprehensive representation of the market state, denoted as S_d , which was then used within an actor-critic framework to learn the optimal trading policy.

The Actor network processes the market state representation S_d and generates a probability distribution function π_θ , as illustrated in Figure 1, which determines the optimal action to take in the current market state (buy, sell, or hold). The objective of the Actor network is to maximize policy entropy while

minimizing the Kullback-Leibler divergence, as described in (9).

$$\mathcal{L}_{\text{act}} = \min_{\theta} \mathbb{E} \left[D_{\text{KL}} \left(\pi_\theta(a_t | s_t) \parallel \frac{\exp(Q_\phi(s_t, \cdot))}{Z_\theta(s_t)} \right) \right] \quad (9)$$

where θ, ϕ , and ψ denote the parameters of the actor network, critic network, and Relation Module, respectively. The trading strategy is executed in the stock trading environment, and the Critic network evaluates the return after executing an action a_t in a given state s_t by calculating $Q(s_t, a_t)$ to assess the quality of the current execution strategy. Optimization is performed using the following (10):

$$\mathcal{L}_{\text{crit}} = \min_{\theta, \phi} \mathbb{E} \left[\frac{1}{2} \left(Q(s_t, a_t) - \hat{Q}(s_t, a_t) \right)^2 \right] \quad (10)$$

The Critic network propagates the analytic gradients of state values back into the Relation Module. Through this joint learning modality, the accuracy of the relationship representation between stocks can be further improved. Meanwhile, the current Decision Module, with maximum entropy as the objective function, not only seeks to maximize the expected return but also encourages greater stochasticity (i.e., entropy) in the strategy. This approach helps the model better adapt to the non-stationary volatility inherent in financial markets.

IV. EXPERIMENTS

A. Experimental Setup

To evaluate the model's performance under various market conditions (such as differing economic environments, trading regulations, and market momentum), we choose the CSI-300 from China and the NASDAQ-100 from the United States as our datasets. We use fundamental and technical stock data, including open price, high price, low price, close price, and volume (OHLCV), which are widely employed in financial market analysis. Technical indicators such as Moving Average Convergence Divergence (MACD), Moving Average (MA), and Relative Strength Index (RSI) will collectively serve as inputs. For the CSI-300, we will select data from 88 stocks over a span of 2663 days, with a training and testing ratio of 7:3. The NASDAQ-100 will follow a similar approach, albeit with 86 stocks chosen. Stocks with incomplete records, such as those with over 2% missing data, are excluded.

For evaluation, we use ARR to normalize returns, ASR to assess return-to-volatility ratios, and MDD to measure peak-to-trough declines, reflecting worst-case losses. These metrics comprehensively assess the model's risk-return profile in volatile markets.

B. Baselines

These comparative methods are classified into traditional and deep learning-based approaches. The Market is a financial index that reflects historical performance and serves as a benchmark. RMR [19] minimizes short-term anomalies, prioritizing underperforming assets—a "follower of losers" approach. Universal Portfolio (UP) selects winners by simulating various portfolio performances and applying weighted

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT MODELS.

Datasets	CSI-300			NASDAQ-100		
	ARR(%)	ASR	MDD(%)	ARR(%)	ASR	MDD(%)
Market	8.24	0.41	51.70	19.09	0.78	43.77
UP	6.12	0.29	74.23	11.71	0.52	66.59
RMR	14.26	0.87	54.32	21.97	0.91	47.91
ELLE	17.03	0.96	65.41	22.8	1.14	54.80
FINRL-DDPG	18.91	1.12	62.14	26.15	1.63	53.32
FINRL-SAC	20.75	1.14	58.37	25.76	1.41	52.36
FD-RLPO	25.28	1.31	56.51	31.8	2.07	48.84

TABLE II
EVALUATION OF THE IMPACT OF INDIVIDUAL BRANCHES IN FD-RLPO ON CSI-300.

Configuration	ARR(%)	ASR	MDD(%)
W/o Relation Module	22.97	1.19	63.50
W/o Prediction Module	23.42	1.24	59.21
Without both	20.91	0.97	72.24
FD-RLPO (our)	25.28	1.31	56.51

averaging. The deep learning-based group includes ELLE [20], which integrates CNNs and RNNs for automated trading in dynamic markets. FinRL-SAC [11] synergizes policy gradient and value function methods to optimize returns and entropy, while FinRL-DDPG [10] excels in continuous action control, enabling dynamic and adaptive portfolio allocation.

C. Evaluation Results

We evaluate the risk-return results of FD-RLPO and other comparative methods in Table I. FD-RLPO outperforms comparative baselines in dynamic market environments, especially in ASR (Sharpe Ratio) and ARR (Annualized Return Rate). On the CSI-300 dataset, FD-RLPO achieves an ARR of 25.28%, a 4.53% improvement over FinRL-SAC, with an ASR of 1.31 and an MDD of 56.51%, outperforming most traditional and deep learning models.

On the NASDAQ-100 dataset, FD-RLPO achieves an ARR of 31.8% and an ASR of 2.07, outperforming other models. The lowest MDD is 43.77% for the market, with traditional methods performing better under static rules and single-market conditions but showing limited focus on returns. Deep learning models generally exhibit higher MDD than traditional methods, likely due to their pursuit of high returns, their tendency to employ nonlinear modeling capabilities, and often adopting aggressive strategies. However, such higher drawdown levels are acceptable when maintaining a relatively high ASR. FD-RLPO's MDD of 48.84% ensures a manageable drawdown, while its optimal ASR demonstrates its ability to achieve high returns while maintaining controlled drawdowns across complex market environments.

D. Ablation Studies

In this section, we conduct ablation studies to analyze the effectiveness of the Prediction and Relation Modules in

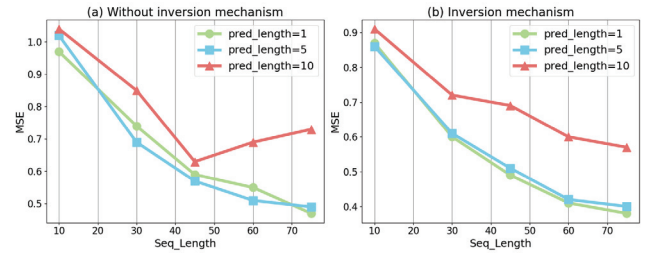


Fig. 2. Performance of Inversion Mechanism with/without Varying Sequence and Prediction Length on CSI-300.

FD-RLPO. As shown in Table II, when only the Relation Module is excluded, MDD rises by 6.99% compared to the full FD-RLPO model, indicating weakened risk control. This reflects the importance of the Relation Module in capturing inter-domain dependencies, which helps stabilize the portfolio by leveraging cross-stock correlations. Meanwhile, removing the Prediction Module reduces ARR to 23.42% and ASR to 1.24, underscoring that the absence of intra-domain predictive capabilities significantly impacts overall returns. Complete removal of both modules leads to a sharp decline across all metrics. These findings suggest that the Relation and Prediction Modules effectively support portfolio optimization in financial markets from both intra- and inter-domain feature extraction perspectives.

E. Evaluation on Inversion Mechanism

To check if the inversion mechanism in the Prediction Module works to lessen the effect of longer prediction lengths on volatility, we set up different sequence lengths (seq_length) to see how well the module could capture the changing intra-domain features in the CSI-300 dataset across different prediction lengths (pred_length). We examined two parameter groups: seq_length = (10, 30, 45, 60, 75) and pred_length = (1, 5, 10), with MSE values for each scenario shown in Figure 2.

Figure 2(a) shows the Prediction Module's performance without the inversion mechanism. As seq_length increases, the MSE curves for pred_length of 1 and 5 exhibit a downward trend, while for pred_length = 10, the MSE rises once seq_length exceeds 45. This suggests that basic Transformers struggle with long sequences, particularly for extended prediction lengths, as the ability to extract intra-sequence information declines.

By contrast, Figure 2(b) presents the results with the inversion mechanism in the Prediction Module. The mechanism effectively mitigates the rise in MSE for pred_length = 10, as seq_length increases, demonstrating its ability to better handle long-term predictions than in Figure 2(a). Moreover, it reduces MSE values across all pred_lengths, highlighting its robustness in stabilizing intra-sequence dynamics. The inversion mechanism alleviates the limitations of traditional Transformer models by re-centering attention on variable-specific features, thereby decreasing intra-sequence volatility and enhancing prediction accuracy. These results emphasize that the inversion mechanism improves the Prediction Module's adaptability to

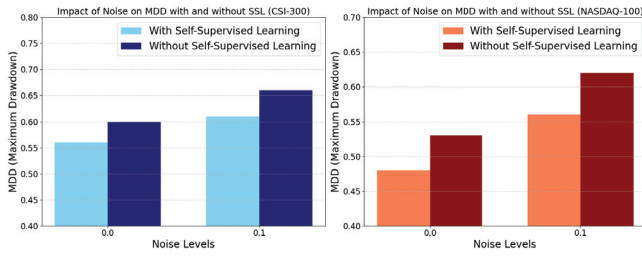


Fig. 3. Impact of Noise on MDD with and without self-supervised learning.

dynamic intra-domain feature changes and ensures stability in the FD-RLPO framework.

F. Robustness Analysis

This experiment evaluates the robustness of FD-RLPO under conditions of data noise, comparing setups with and without self-supervised learning (SSL). We induced Gaussian noise into the raw data (with zero mean and varying standard deviations). The level of noise is measured by the standard deviation. Two scenarios were considered: no noise ($\sigma = 0.0$) and noise with a standard deviation of 0.1 ($\sigma = 0.1$). The results are presented in Figure 3.

We observe that FD-RLPO with SSL effectively handles data noise. Across the CSI-300 and NASDAQ-100 datasets, SSL consistently outperforms the non-SSL configuration, regardless of noise presence. In noise-free conditions, SSL reduces MDD to 56% on CSI-300 (4% lower than non-SSL) and 48% on NASDAQ-100, outperforming the non-SSL setup. Under noisy conditions, SSL's advantage becomes more pronounced, as noise increases MDD in the non-SSL setup by 8% on NASDAQ-100, highlighting SSL's effectiveness in mitigating MDD. This demonstrates the generalizability and robustness of SSL's approach across markets. This phenomenon is mainly attributed to the fact that the reconstruction mechanism of SSL better preserves the key patterns of the data under noise. The assembly of SSL enables FD-RLPO to effectively mitigate the interference of noise on the feature representations and make the model more robust.

V. CONCLUSIONS

In this paper, we propose FD-RLPO, a feature-domain-based reinforcement learning model designed to handle market volatility through three key modules: Prediction, Relation, and Decision. The Prediction Module employs an inversion mechanism within the Transformer architecture to extract intra-domain information by constructing variable-centered representations, ensuring stability over long sequences. The Relation Module utilizes self-supervised learning to reconstruct asset relationships, capturing inter-domain stock information and mitigating data noise. Outputs from both modules are integrated via multi-attention into a comprehensive representation, which the Decision Module uses to generate a decision function for portfolio optimization. FD-RLPO provides a complementary framework by concurrently collecting both intra-domain and inter-domain features, resulting in better returns

in highly volatile financial environments. Comparative experiments on domestic and international markets demonstrate its superior performance in ARR and ASR, while ablation studies confirm the effectiveness of its core components.

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